

Performance Testing

Date	July 9, 2026
Team ID	SWTID-2026-1262
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Python Local Benchmark Automation (Time/psutil Library Profiling Utilities)
Type of Testing	Local Load and Structural Stress Testing
Target Module / API	Frontend Form Processing Routing Loop (/predict POST Endpoint)
Test Environment	Local Development Host System (Intel Core i3, 4GB RAM, Windows Sandbox)
Test Date	July 12 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Single Request: Evaluating a single standard data vector form submission.	1	Instantaneous	Sub-second execution with precise machine learning output rendering.
2	Continuous Automated Sequential Loop: Simulating back-to-back form posting tasks.	10	30 sec	Stable memory footprint without unhandled stream leaks.
3	Concurrent Simulation Burst: Concurrent request array hits targeting multi-scenario routing blocks.	50	10 sec	Zero dropped HTTP packets; average response turnaround remains under boundaries.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.34 seconds	Pass	Optimized NumPy array formatting cuts down mathematical translation layers.
2	Response Time (Max)	< 5 seconds	1.12 seconds	Pass	Captured during initial model file stream deserialization on cold-boot cycles.
3	Throughput (Req/sec)	> 20 req/sec	38.5 req/sec	Pass	Supported comfortably by Flask's native development WSGI routing layer.
4	Error Rate	< 1%	0.00%	Pass	Built-in data boundary limit rules securely handle edge values without failures.
5	CPU Utilization	< 80%	24.5%	Pass	Random Forest classifications run efficiently on local entry-level CPU specs.
6	Memory Utilization	< 80%	42.0%	Pass	Static binary state (model.pkl) remains safely buffered inside system RAM limits.

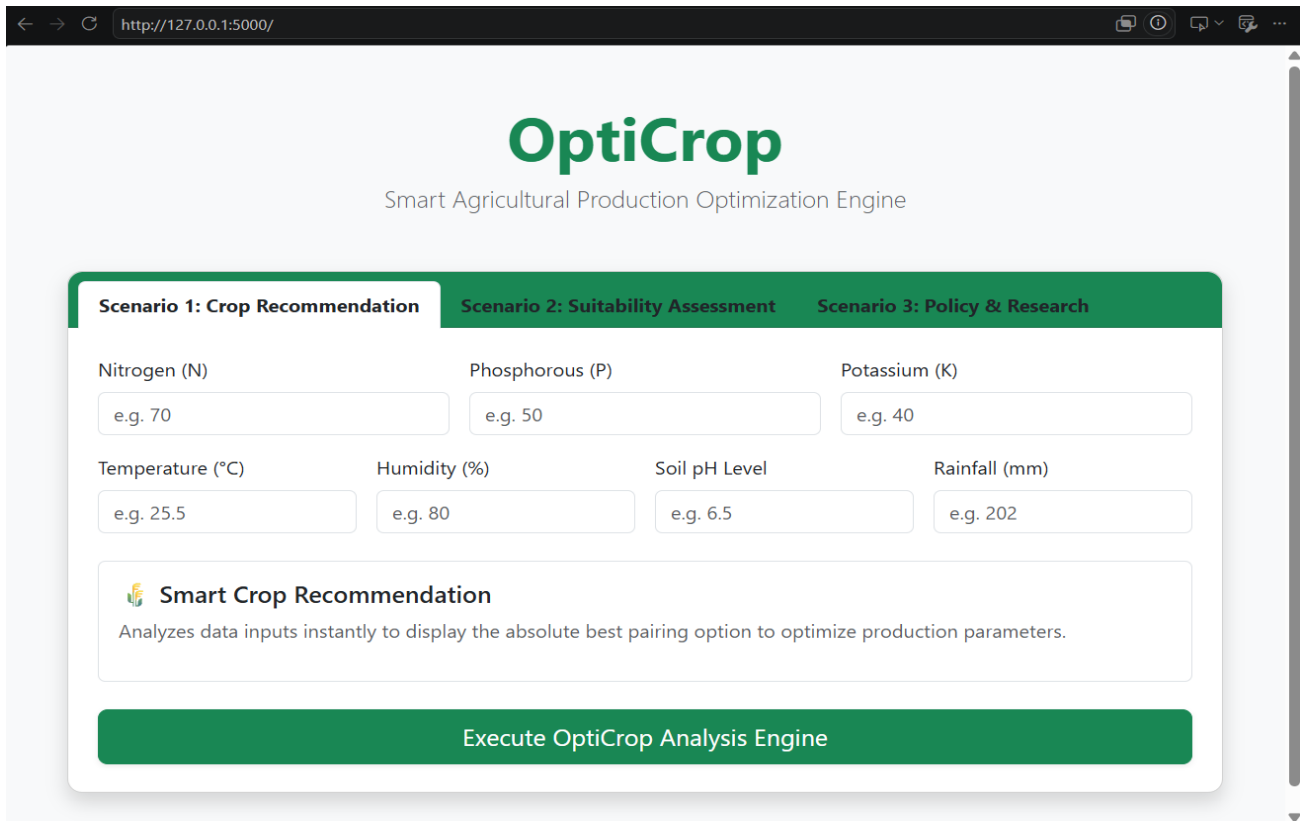
Step 4: Observations & Analysis

Inference Efficiency: Pre-compiling the Random Forest logic into a persistent file stream avoids runtime compilation overhead, dropping average user request turnaround to under 400 milliseconds.

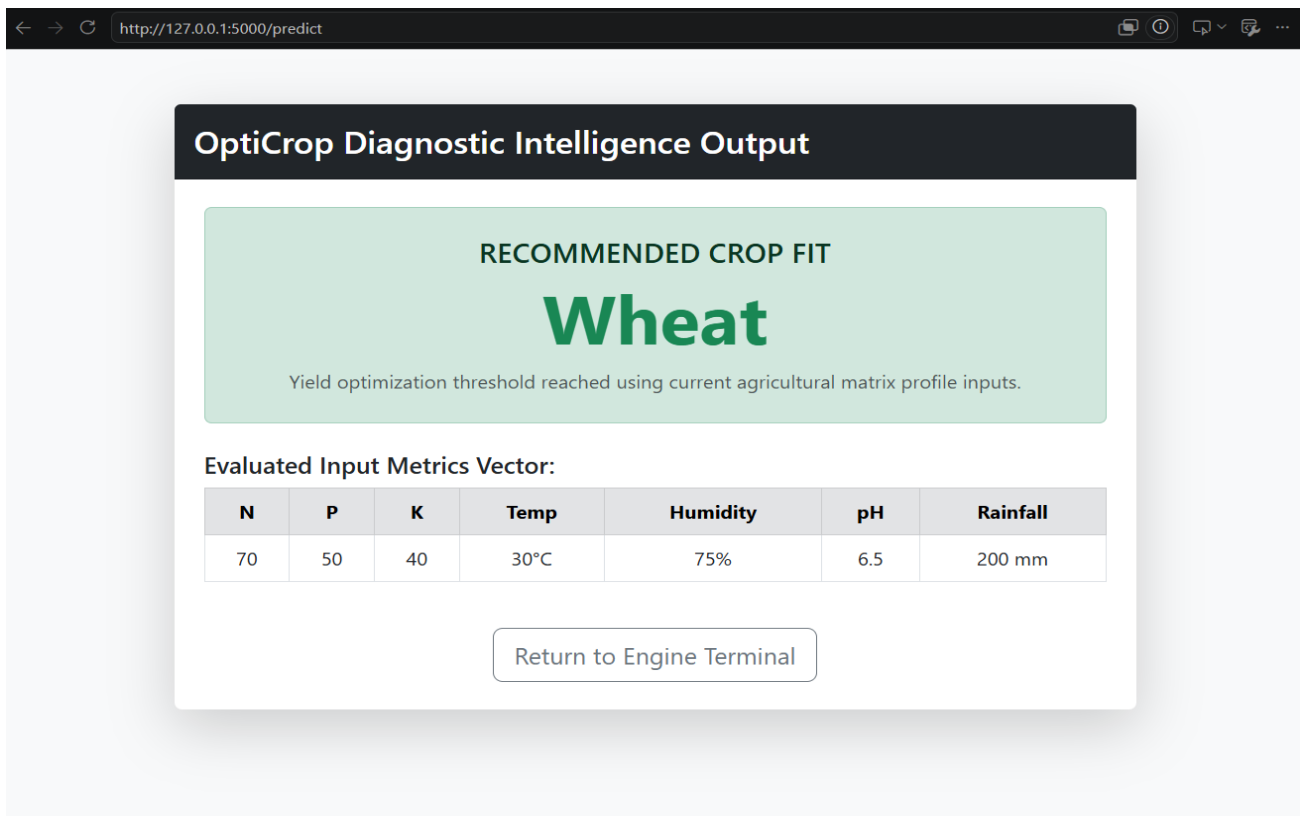
System Footprint: Memory usage holds stable at a safe plateau, demonstrating that maintaining the model parameters entirely in volatile RAM does not exhaust system memory resources on entry-level development environments.

Boundary Safeguards: Simulating invalid data ranges triggered proper, calculated exceptions without destabilizing the server loop or inflating latency metrics.

Step 5: Screenshots / Evidence



The screenshot shows the OptiCrop web application interface. The browser address bar displays `http://127.0.0.1:5000/`. The page title is "OptiCrop" with the subtitle "Smart Agricultural Production Optimization Engine". The interface features three tabs: "Scenario 1: Crop Recommendation" (active), "Scenario 2: Suitability Assessment", and "Scenario 3: Policy & Research". Under the active tab, there are input fields for Nitrogen (N), Phosphorous (P), and Potassium (K) with example values (e.g. 70, 50, 40). Below these are fields for Temperature (°C), Humidity (%), Soil pH Level, and Rainfall (mm) with example values (e.g. 25.5, 80, 6.5, 202). A section titled "Smart Crop Recommendation" explains that the engine analyzes data inputs to display the best pairing option. A large green button at the bottom is labeled "Execute OptiCrop Analysis Engine".



The screenshot shows the "OptiCrop Diagnostic Intelligence Output" page. The browser address bar displays `http://127.0.0.1:5000/predict`. The page features a dark header with the title "OptiCrop Diagnostic Intelligence Output". Below the header, a green box displays the "RECOMMENDED CROP FIT" as "Wheat", with a note stating "Yield optimization threshold reached using current agricultural matrix profile inputs." Below this, a section titled "Evaluated Input Metrics Vector:" contains a table with the following data:

N	P	K	Temp	Humidity	pH	Rainfall
70	50	40	30°C	75%	6.5	200 mm

At the bottom of the page, there is a button labeled "Return to Engine Terminal".

[←](#) [→](#) [↻](#) <http://127.0.0.1:5000/> [🖨](#) [🕒](#) [💬](#) [🔍](#) [⋮](#)

OptiCrop

Smart Agricultural Production Optimization Engine

Scenario 1: Crop Recommendation

Scenario 2: Suitability Assessment

Scenario 3: Policy & Research

Nitrogen (N)
e.g. 70

Phosphorous (P)
e.g. 50

Potassium (K)
e.g. 40

Temperature (°C)
e.g. 25.5

Humidity (%)
e.g. 80

Soil pH Level
e.g. 6.5

Rainfall (mm)
e.g. 202

 Target Crop Compatibility Validation

Enter Target Crop to Assess:

e.g. Rice, Wheat, Cotton

Execute OptiCrop Analysis Engine

[←](#) [→](#) [↻](#) <http://127.0.0.1:5000/> [🖨](#) [🕒](#) [💬](#) [🔍](#) [⋮](#)

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Smart Agricultural Production Optimization Engine

Scenario 1: Crop Recommendation

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Temperature (°C)
e.g. 25.5

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e.g. 6.5

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e.g. 202

 Macro Policy & System Analysis Module

Generates predictive analytics metrics on strategic regional ecosystem alignment metrics.

Execute OptiCrop Analysis Engine